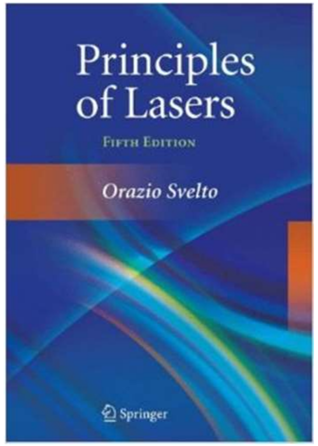
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch6-2
Control & Improvement of Laser Properties

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Chapter 6: Control & Improvement of Laser Performance

❖ 6.3 *Q*-Switch

- Working principle of *Q*-switching laser
- *Q*-switching methods

❖ 6.4 Mode Locking: ultrashort pulse (ps-fs)

- Outline
- Mode locking principle

6.3 *Q*-Switching

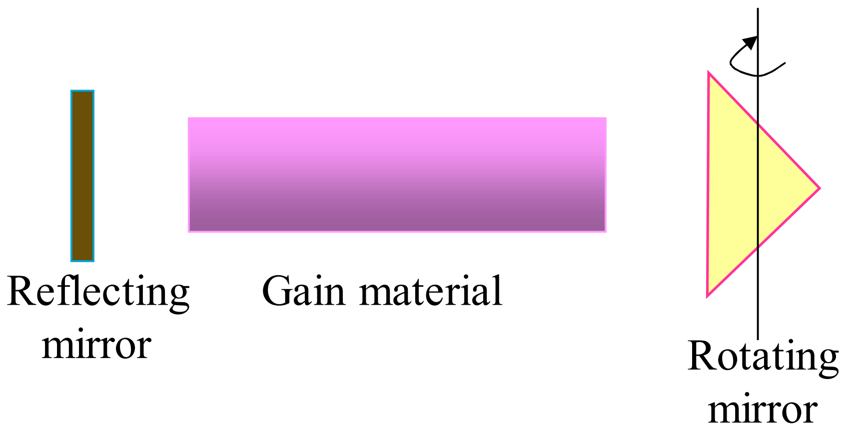
Rotating Mirror: reflection loss

Electro-Optic: reflection loss

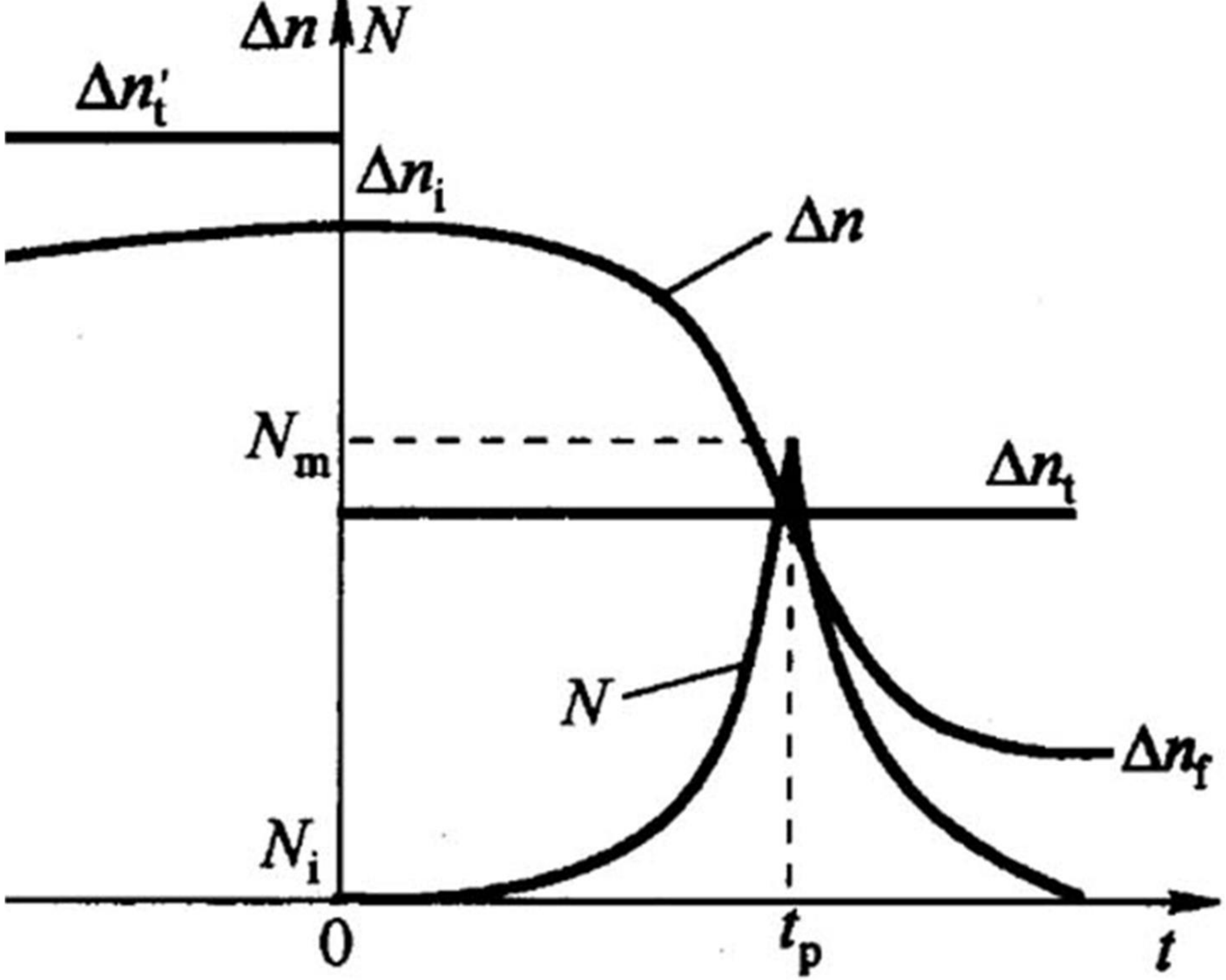
Acousto-opitc: diffraction loss

Saturable absorber: absorption loss

- **Active** *Q*-Switching: External drive source to adjust intracavity loss
  Examples: electro-optic and acousto-optic
- **Passive** *Q*-Switching: Intracavity intensity to adjust loss
  Example: saturable absorber dyes
- **Slow switch** Switch time > Pulse build-up time
  Example: Rotating mirror
- **Fast switch** Switch time < Pulse build-up time

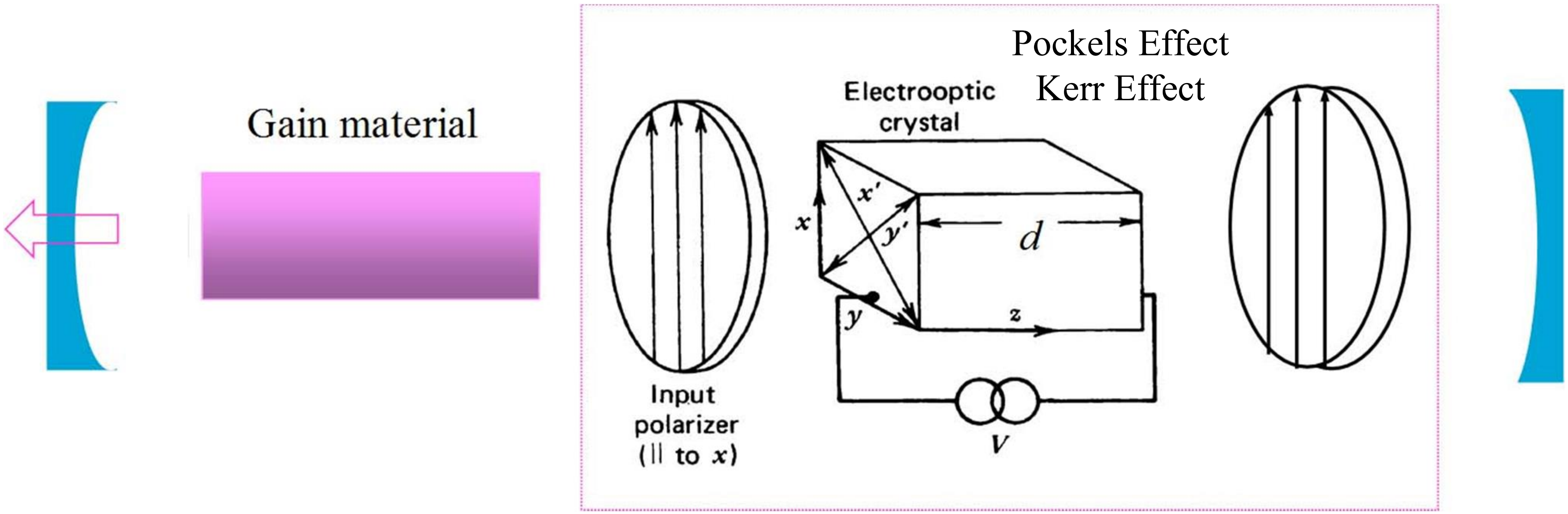


❖ Working principle of Q-switching laser

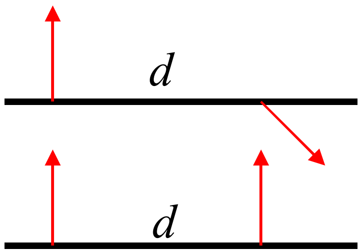


❖ *Q*-switching methods

➢ Electro-optic *Q*-switching



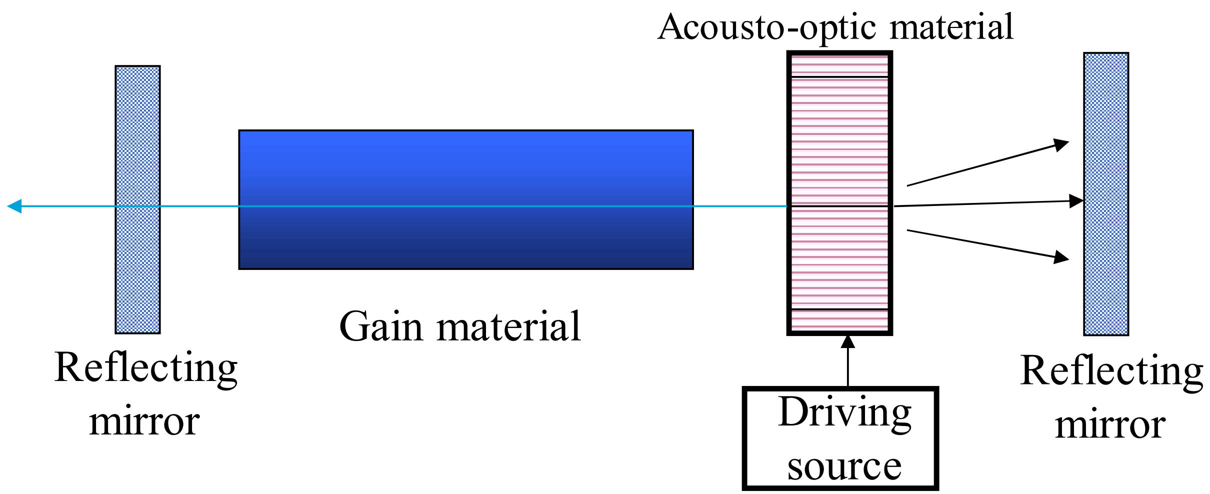
$$\Delta\Phi = \frac{2\pi\nu d}{c}\left(\eta'_y - \eta'_x\right) = \frac{2\pi\nu\eta_0^3\gamma_{63}}{c}Ed = \frac{2\pi\nu\eta_0^3\gamma_{63}}{c}V$$



$V = V_\pi$ — *Q* switch off — Δn accumulate

$V = 0$ — *Q* switch on — Oscillate to form a giant pulse

➢ Acousto-optic Q-switching: low gain CW laser

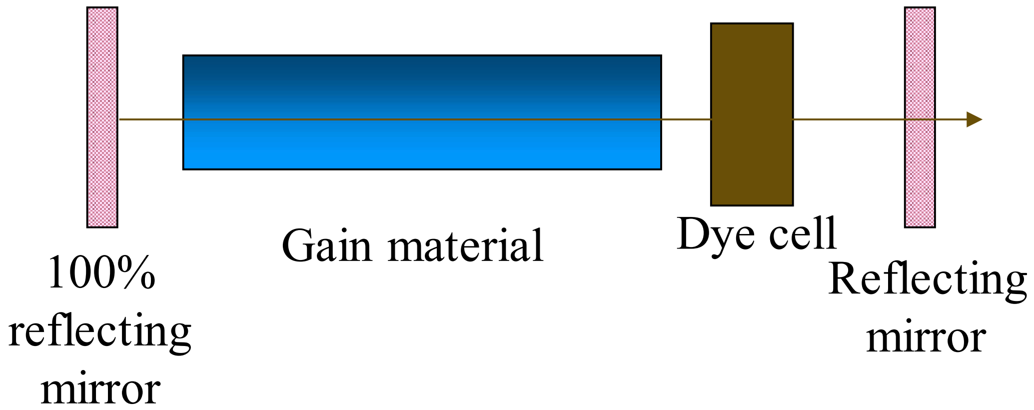


Bragg diffraction

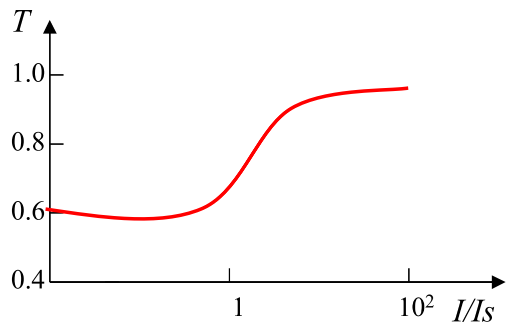
$$\frac{I_1}{I_i} = \sin^2\left(\frac{\Delta\varphi}{2}\right)$$

$$\Delta\varphi = \frac{\pi}{\lambda}\left(2\frac{d}{H}MP\right)^{1/2}$$

➢ Saturable absorbing dye (passive Q-switching)



$$I_s' = \frac{h\nu}{2\sigma_{12}\tau_2}$$



6.4 Mode Locking: ultrashort pulse (ps-fs)

❖ Outline

Output characteristics of a free running (non-mode-locked) multimode laser

The expression of electric field for each longitudinal mode

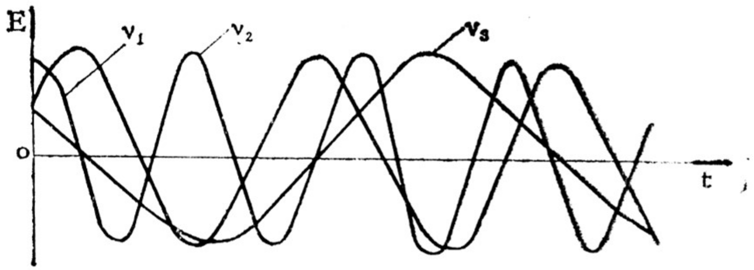
$$E_q(t) = E_q \cos\left(\omega_q t + \varphi_q\right)$$

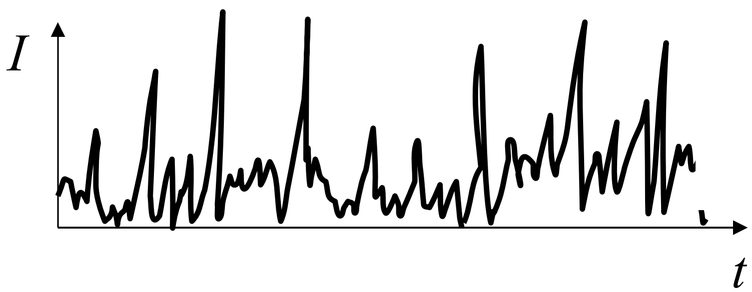
Frequency interval between adjacent longitudinal modes

$$\Delta \nu_q = \frac{\nu}{2L} \qquad \Delta \omega_q = \frac{\pi \nu}{L} = \Omega$$

Total output

$$E(t) = \sum_q E_q e^{i\left(\omega_q t + \varphi_q\right)}$$





Different build-up time for each longitudinal mode

Different E_q, ω_q, and φ_q

Output: random superposition of multiple longitudinal modes

❖ Mode locking principle

Multi longitudinal mode phase locking $\varphi_{q+1} - \varphi_q = a$

Mode locking mechanism: keep the same initial phase for the longitudinal modes in the cavity or to have a definite phase relation between the longitudinal modes, then the output is equal interval short pulse sequence.

Example: Superposition of 3 longitudinal modes after locking

ν_1 , ν_2 , ν_3

$\nu_2 = 2\nu_1$, $\nu_3 = 3\nu_1$

$E_1 = E_0 \cos(2\pi\nu_1 t)$

$E_1 = E_2 = E_3 = E_0$

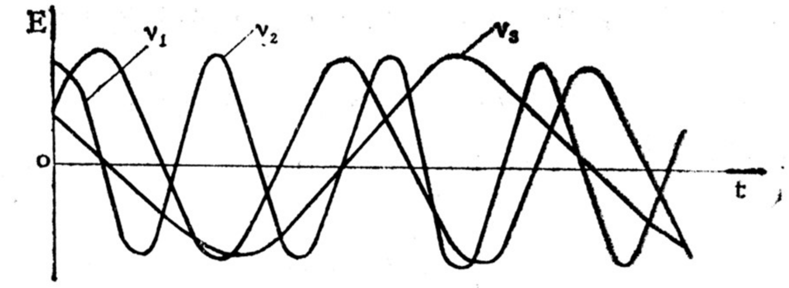
$\varphi_1 = \varphi_2 = \varphi_3 = 0$

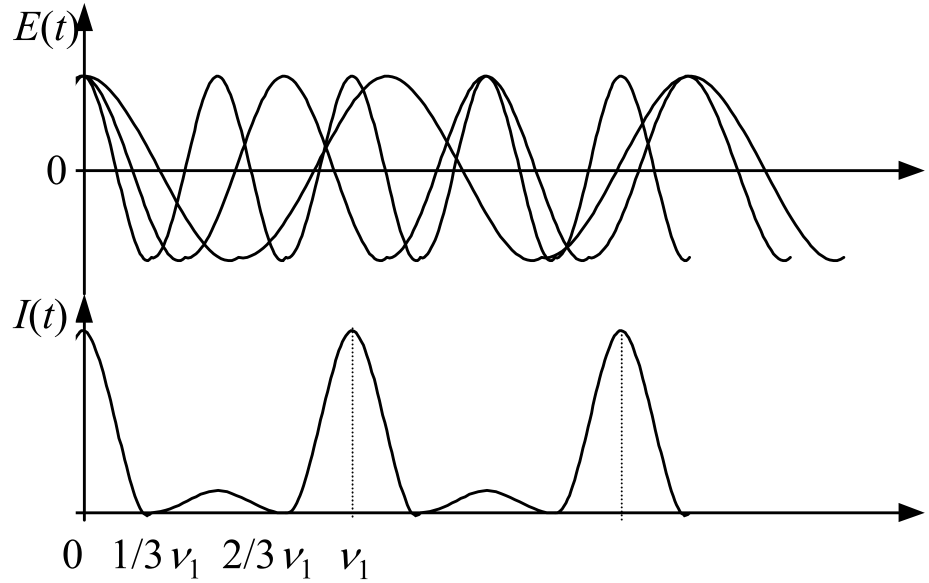
$E_2 = E_0 \cos(4\pi\nu_1 t)$

$E_3 = E_0 \cos(6\pi\nu_1 t)$

(equal amplitude)

(initial phase=0)





$$E_1 = E_0 \cos(2\pi\nu_1 t)$$
$$E_2 = E_0 \cos(4\pi\nu_1 t)$$
$$E_3 = E_0 \cos(6\pi\nu_1 t)$$

$t=0$	$E_1=E_2=E_3=E_0$	$E=3E_0$	$E^2=9E_0^2$	Maximum
$t=1/3\nu_1$	$E_1=E_2=-E_0/2,\ E_3=E_0$	$E=0$		Minimum
$t=2/3\nu_1$	$E_1=E_2=-E_0/2,\ E_3=E_0$	$E=0$		Minimum
$t=1/\nu_1$	$E_1=E_2=E_3=E_0$	$E=3E_0$	$E^2=9E_0^2$	Maximum

Discussion: Assume $2N+1$ mode oscillation, equal amplitude (E_0)

- Phase difference between adjacent longitudinal modes $\quad \varphi_q - \varphi_{q-1} = \beta \qquad \varphi_q = \varphi_0 + q\beta$
- Angular frequency interval between adjacent longitudinal modes $\quad \Delta\omega_q = \pi c/L = \Omega$

q	$-N$	$-(N-1)$	…	-2	-1	0	1	2	…	$(N-1)$	N
ω_q	$\omega_0-N\Omega$	$\omega_0-(N-1)\Omega$			$\omega_0-\Omega$	ω_0	$\omega_0+\Omega$			$\omega_0+(N-1)\Omega$	$\omega_0+N\Omega$
φ_q	$-N\beta$	$-(N-1)\beta$			$-\beta$	0	β			$(N-1)\beta$	$N\beta$

When locked $\quad E(t) = \sum_{q=-N}^{N} E_0 \cos\left[\left(\omega_0 + q\Omega\right)t + q\beta\right] = E_0 \cos\omega_0 t \sum_{q=-N}^{N} \cos q\left(\Omega t + \beta\right)$

Amplitude

$$A(t) = E_0 \frac{\sin\frac{1}{2}(2N+1)(\Omega t+\beta)}{\sin\frac{1}{2}(\Omega t+\beta)}$$

$$E(t) = E_0 \underline{\cos\omega_0 t} \frac{\sin\frac{1}{2}(2N+1)(\Omega t+\beta)}{\sin\frac{1}{2}(\Omega t+\beta)}$$

Amplitude modulation envelope

❖ Mode-locked pulse characteristics

➢ Peak power I_m (Max intensity)

$$I(t) \propto A^2(t) = E_0^2 \frac{\sin^2 \frac{1}{2}(2N+1)(\Omega t + \beta)}{\sin^2 \frac{1}{2}(\Omega t + \beta)} \Rightarrow$$

Un-locked $I \propto (2N+1) E_0^2$

$(\Omega t + \beta) \to 2m\pi \qquad (m = 0, 1 \ldots)$

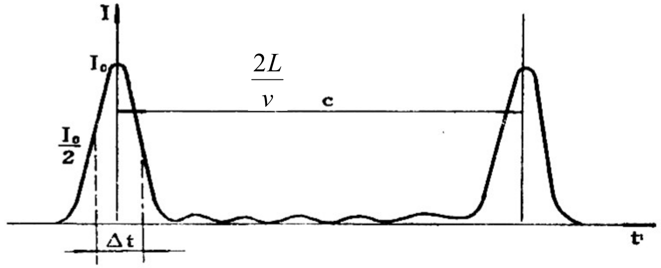
$\boxed{I_m = (2N+1)^2 E_0^2}$ have maximum value

- ✓ The peak power has **(2*N*+1)** times difference before and after mode-locking
- ✓ The number of modes is beneficial to improve the peak power of the mode-locked pulse
- ✓ Ways of increasing oscillation modes: $L\uparrow \to \Delta\nu_q\downarrow$ and $\Delta\nu_F\uparrow$ (broad fluorescent line)

➢ Mode-locked pulse interval

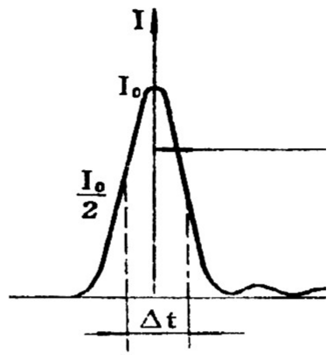
Interval between adjacent pulse maxima (T_0)

$$T_0 = \frac{2\pi}{\Omega} = \frac{2L'}{c} = \frac{1}{\Delta\nu_q}$$



➤ Pulse width (τ)

$$\tau = \frac{2\pi}{(2N+1)\Omega} = \frac{2L'}{c}\frac{1}{2N+1} = \frac{1}{\Delta\nu_q(2N+1)} = \frac{1}{\Delta\nu_{osc}} \approx \frac{1}{\Delta\nu_F}$$

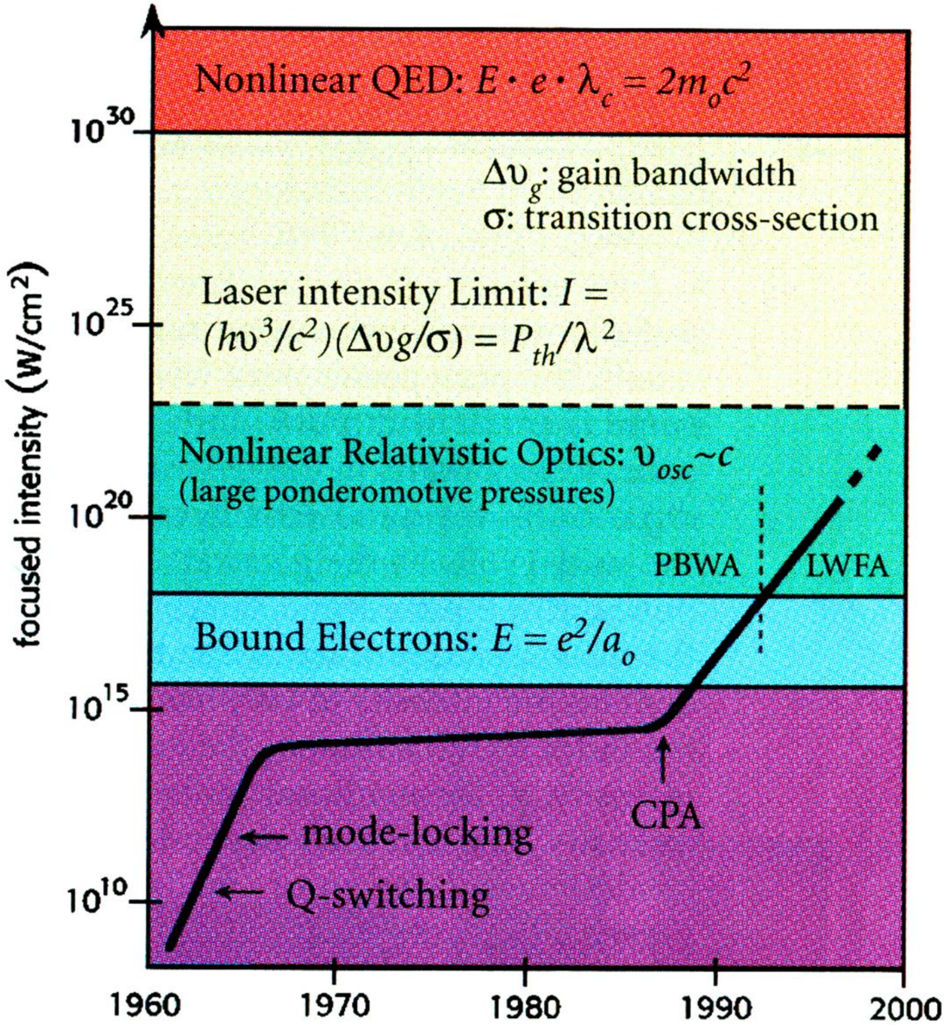


Estimate mode-locked pulse width by $\Delta\nu_F$

Neodymium glass $\Delta\nu_F$=7.5×10^{12} (ps)

Lasers	**$\Delta\nu_F$ (s^{-1})**	**$1/\Delta\nu_F$ (s)**	**Measured pulse width (τ)**
He-Ne	1.5×10^{9}	6.66×10^{-10}	～6×10^{-10}
Nd:YAG	1.95×10^{11}	5.2×10^{-12}	7.6×10^{-12}
Nd:Glass	7.5×10^{12}	1.33×10^{-13}	4×10^{-13}
R6G	5×10^{12}~3×10^{13}	2×10^{-13}～3×10^{-14}	3×10^{-14}
GaAlAs (0.85 μm)	10^{13}	10^{-13}	(0.5～30)$\times10^{-12}$
InGaAsP (1.55 μm)	10^{12}~10^{13}	10^{-12}～10^{-13}	(4～50)$\times10^{-12}$

What to do with such high intensities



Maximum intensity on target

$$I_{\text{peak}} = \frac{E}{A\,\delta t}$$

Pulse energy

Beam size

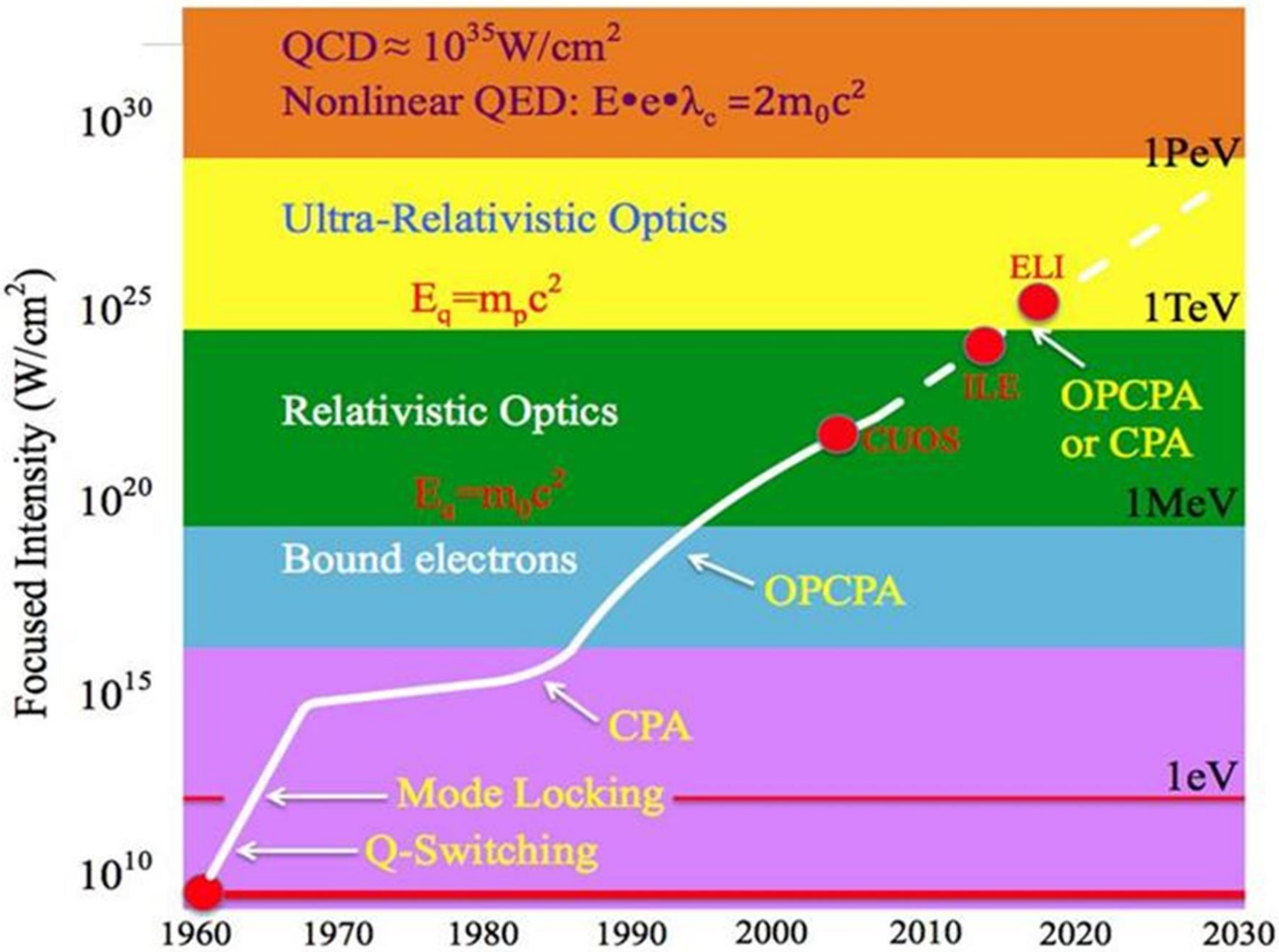
Pulse width

Maximum average power at the detector

$$P_{\text{ave}} = E\,r$$

Pulse energy

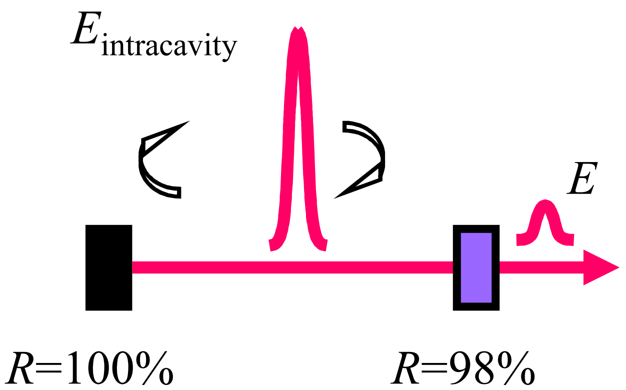
Rep rate


QCD ≈ 10^{35}W/cm^2
Nonlinear QED: E•e•λ_c =2m_0c^2
Ultra-Relativistic Optics
$E_q=m_pc^2$
Relativistic Optics
$E_q=m_0c^2$
Bound electrons
1PeV
1TeV
1MeV
1eV
ELI
ILE
CUOS
OPCPA or CPA
OPCPA
CPA
Mode Locking
Q-Switching
Focused Intensity (W/cm^2)
10^{30}
10^{25}
10^{20}
10^{15}
10^{10}
1960
1970
1980
1990
2000
2010
2020
2030

Cavity Dumping

Before we consider amplification, recall that the intracavity pulse energy is ~50 times the output pulse energy. So we have more pulse energy.

How can we get at it?



$$E = T_{\text{output}}\, E_{\text{intracavity}}$$

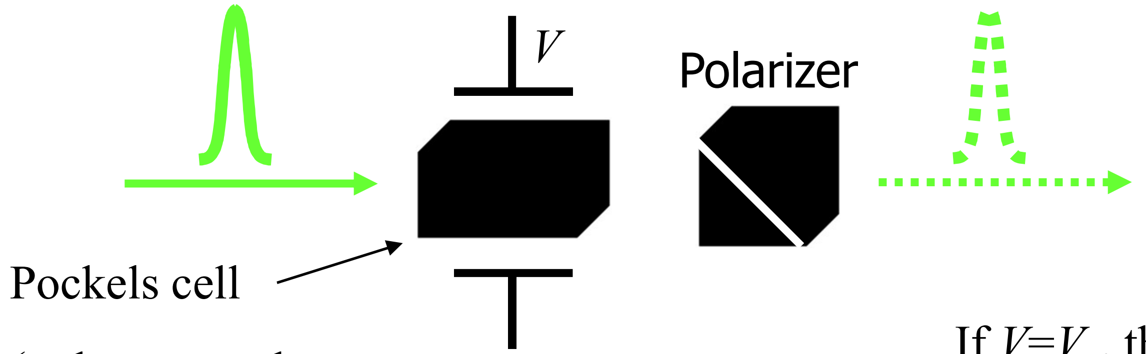
Transmission of output coupler: ~2%

What if we instead used two high reflectors, let the pulse energy build up, and then switch out the pulse. This is the opposite of *Q*-switching: it involves switching from minimum to maximum loss, and it's called "Cavity Dumping".

Cavity dumping: the Pockels cell

A Pockels cell is a device that can switch a pulse (in and) out of a resonator. It's used in Q-switches and cavity dumpers.

A voltage (a few kV) can turn a crystal into a half- or quarter-wave plate.



(voltage may be transverse or longitudinal)

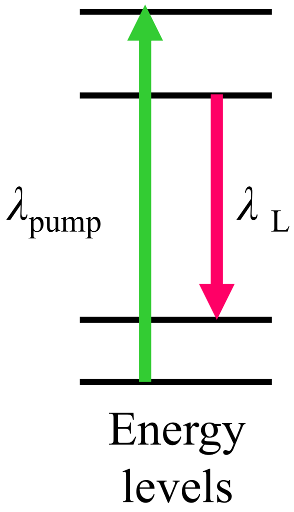
If V=0, the pulse polarization doesn't change.

If V=V_p, the pulse polarization switches to its orthogonal state.

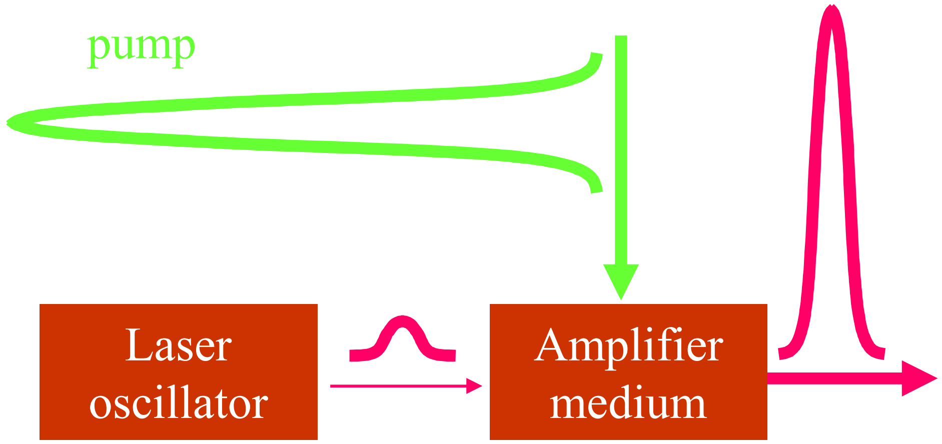
Abruptly switching a Pockels cell allows us to extract a pulse from a cavity. This allows us to achieve ~100 times the pulse energy at 1/100 the repetition rate (i.e., 100 nJ at 1 MHz).

Amplification of Laser Pulses, in General

Very simply, a powerful laser pulse at one color pumps an amplifier medium, creating an inversion, which amplifies another pulse.



λ_{pump} (λ_{pump}/λ_L)

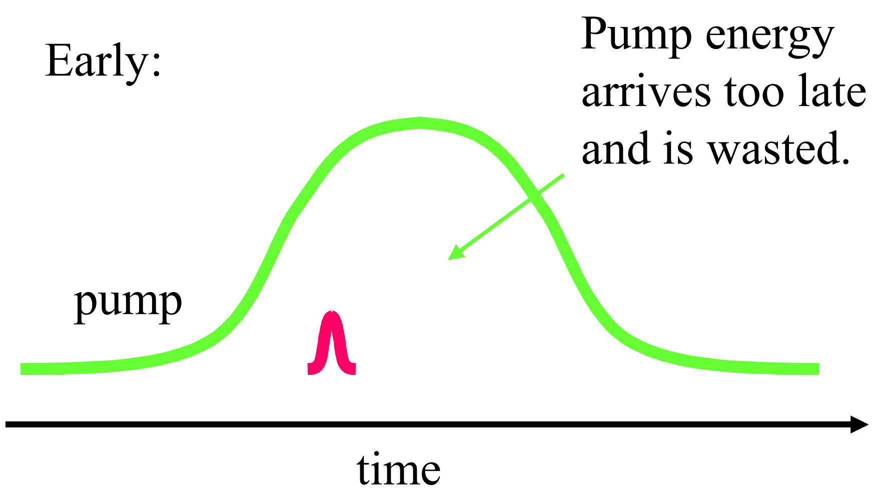


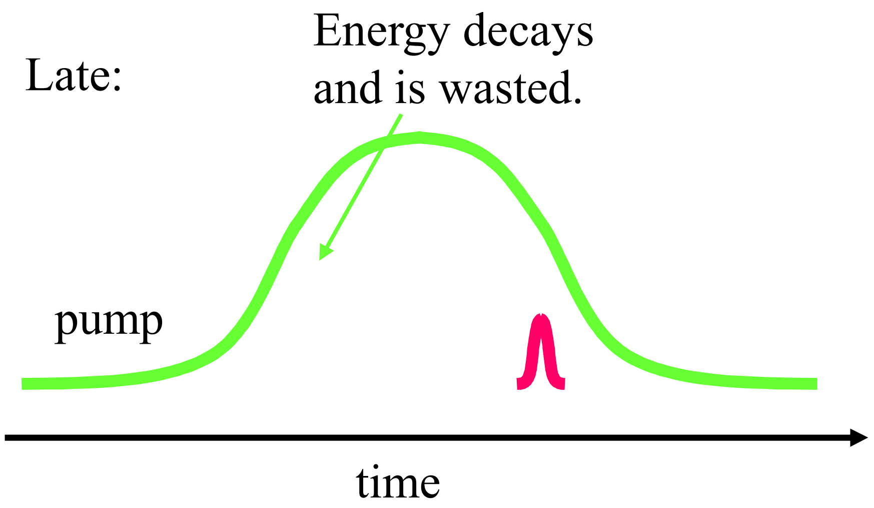
Nanosecond-pulse laser amplifiers pumped by other ns lasers are commonplace.

Another problem with amplifying ultrashort laser pulses…

Another issue is that the ultrashort pulse is so much shorter than the (ns or ms) pump pulse that supplies the energy for amplification.

So should the ultrashort pulse arrive early or late?



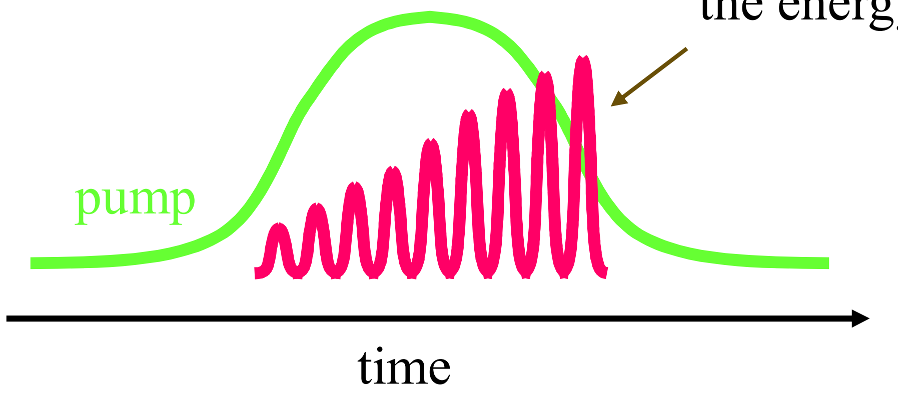


In both cases, pump pulse energy is wasted, and amplification is poor.

So we need many passes

All ultrashort-pulse amplifiers are multi-pass.

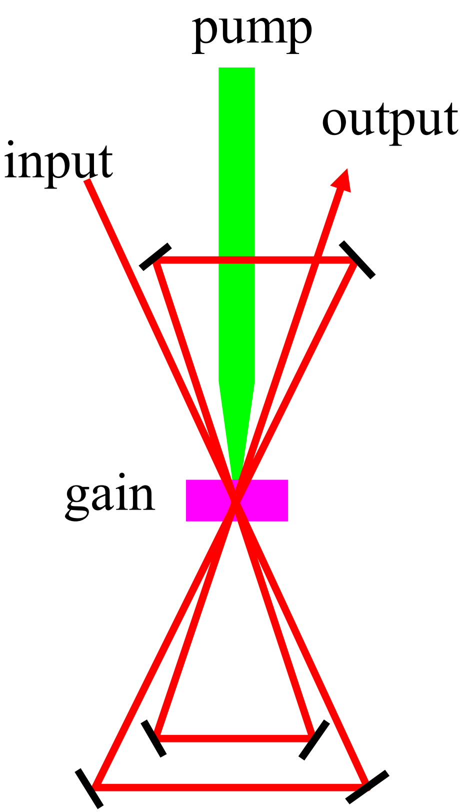
The ultrashort pulse returns many times to eventually extract most of the energy.



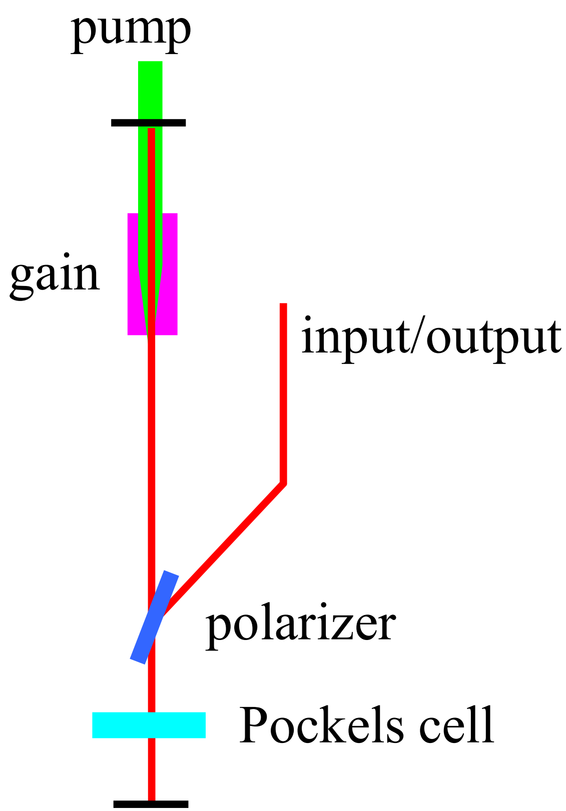
This approach achieves much greater efficiency.

Two main amplification methods

Multi-pass amplifier



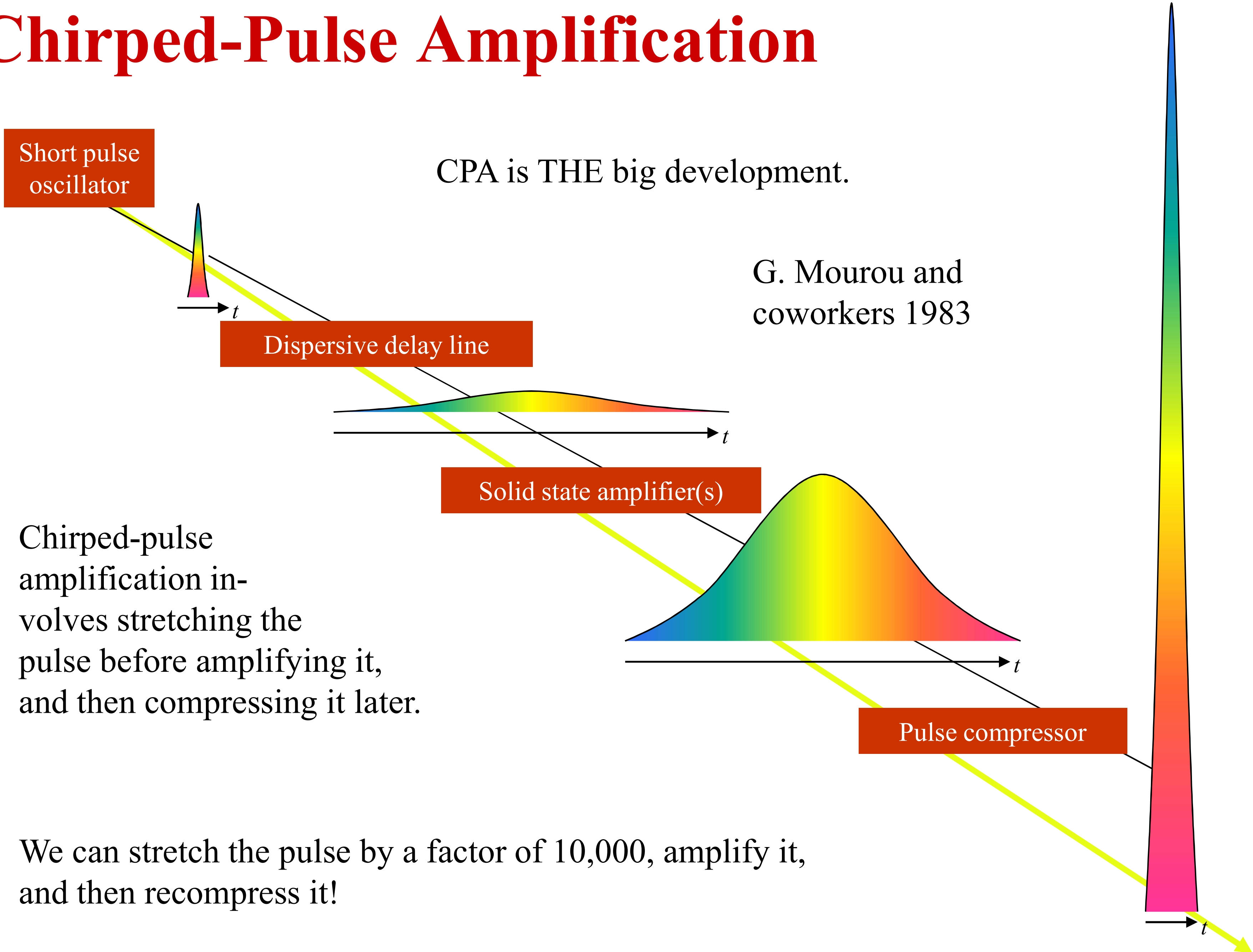
Regenerative amplifier



Chirped-Pulse Amplification

CPA is THE big development.

G. Mourou and coworkers 1983



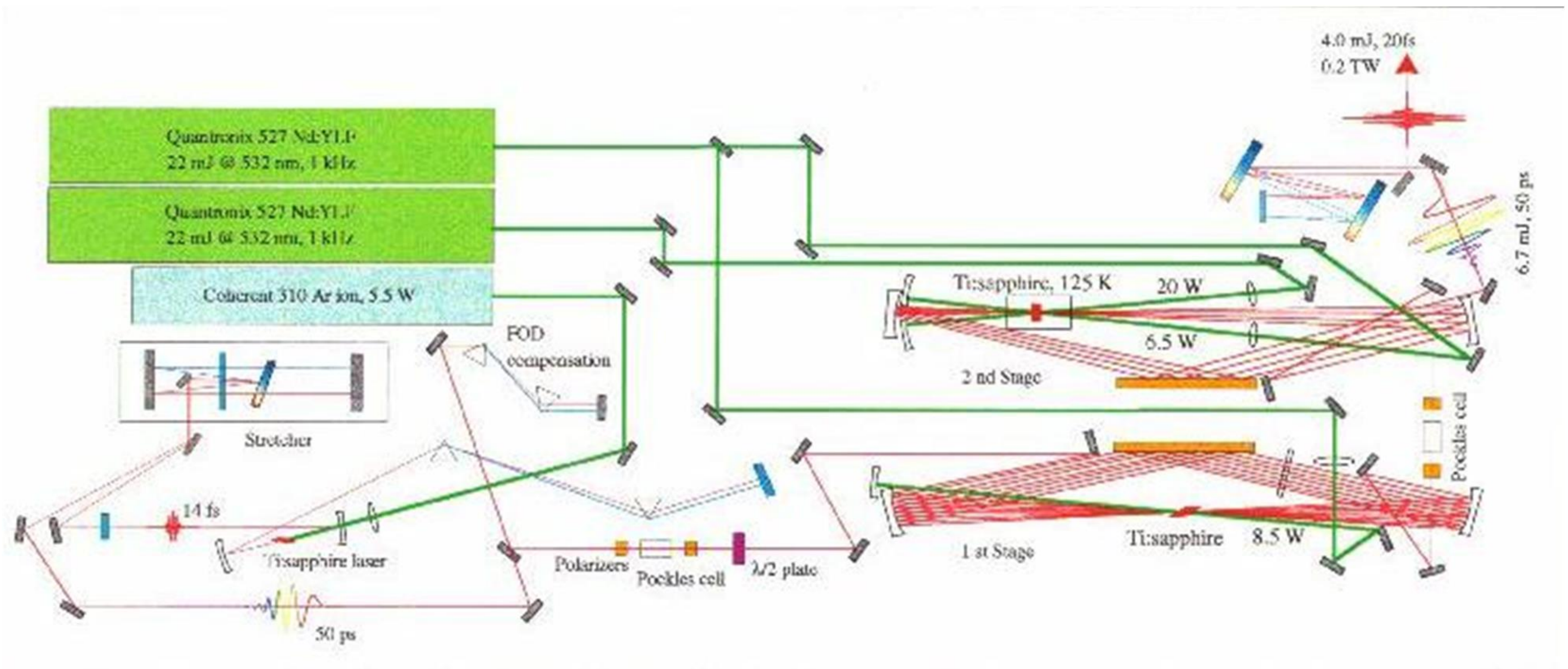
Chirped-pulse amplification involves stretching the pulse before amplifying it, and then compressing it later.

We can stretch the pulse by a factor of 10,000, amplify it, and then recompress it!

Multiple-stage multi-pass amplifiers

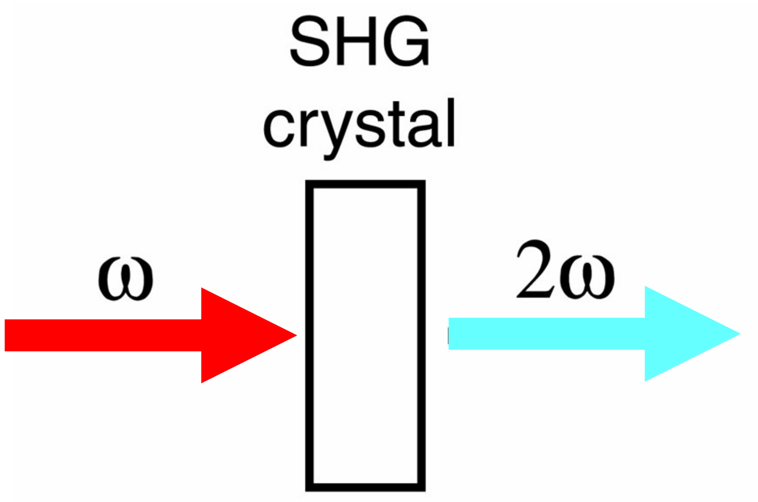
4 mJ, 20 fs pulse length

0.2 TW



1 kHz Multi-pass system at the
University of Colorado (Murnane and Kapteyn)

Phase-matching in second-harmonic generation



How does phase-matching affect SHG? It's a major effect, another important reason you just don't see SHG—or any other nonlinear-optical effects—every day.

First demonstration of second-harmonic generation

P.A. Franken, et al, Physical Review Letters 7, p. 118 (1961)

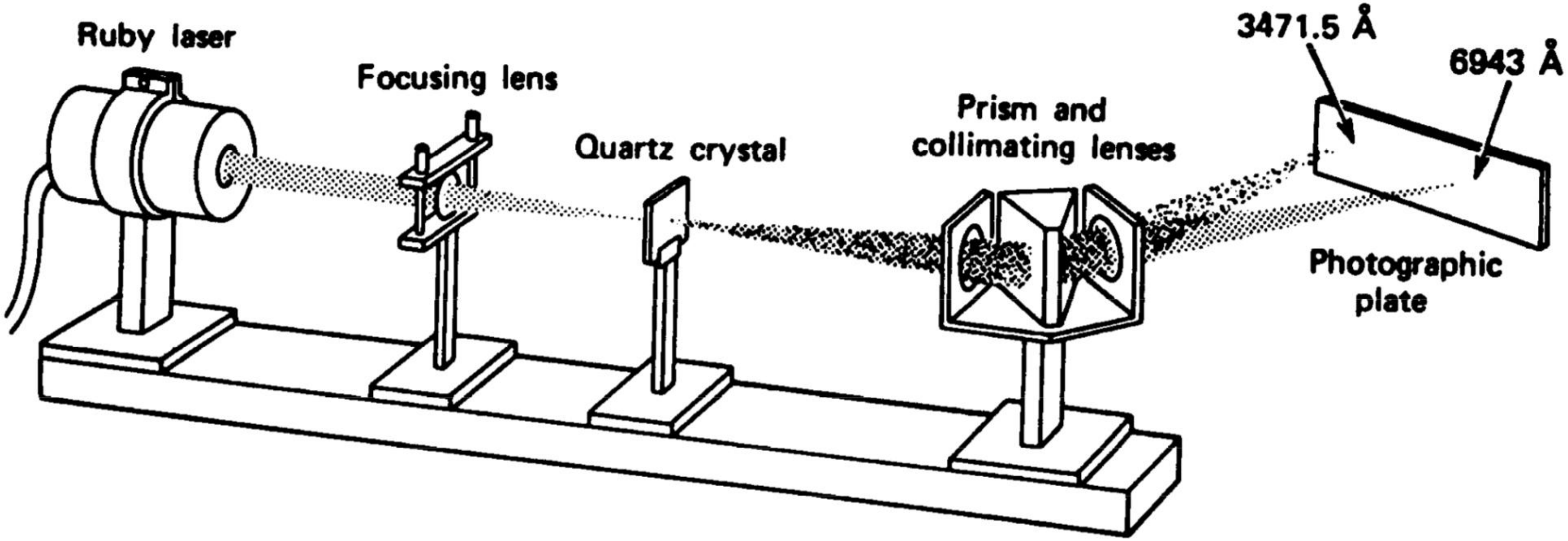


Figure 12.1. Arrangement used in the first experimental demonstration of second-harmonic generation [1]. A ruby-laser beam at $\lambda = 0.694\ \mu m$ is focused on a quartz crystal, causing the generation of a (weak) beam at $\frac{1}{2}\lambda = 0.347\ \mu m$. The two beams are then separated by a prism and detected on a photographic plate.

The second-harmonic beam was very weak because the process was not phase-matched.

First demonstration of SHG: the data

The actual published results…

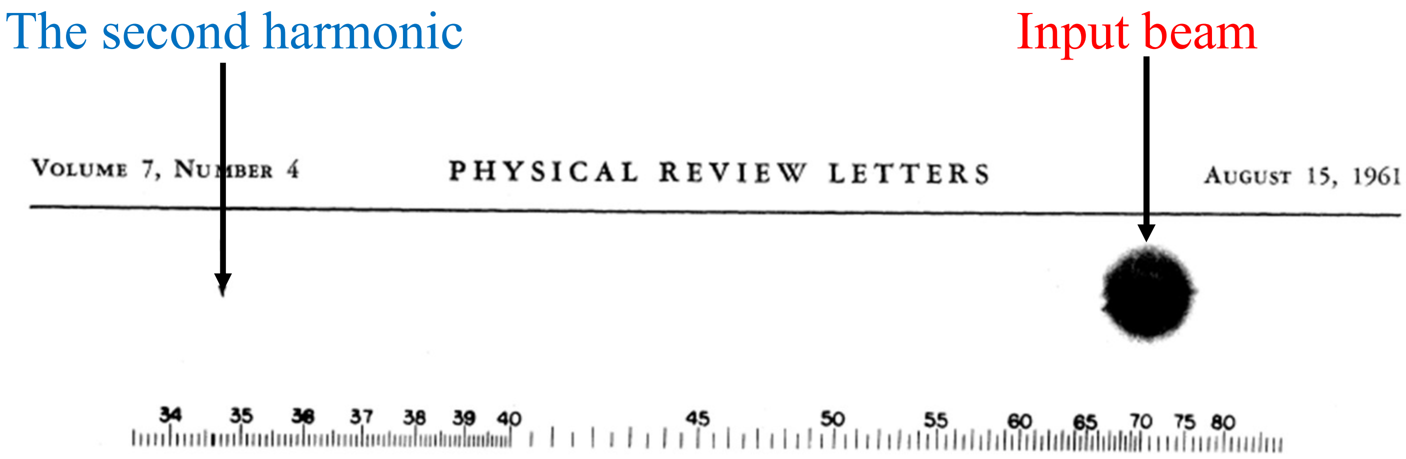


FIG. 1. A direct reproduction of the first plate in which there was an indication of second harmonic. The wavelength scale is in units of 100 A. The arrow at 3472 A indicates the small but dense image produced by the second harmonic. The image of the primary beam at 6943 A is very large due to halation.

Note that the very weak spot due to the second harmonic is missing. It was removed by an overzealous Physical Review Letters editor, who thought it was a speck of dirt.

Bloembergen, Nobel Physics 1980